

24/06/26

Signal Processing

- Recap: Causal and stable systems ✓
- Linear time-invariant (LTI) systems
 - Linearity + time-invariance ✓
 - Convolution sum ✓
 - Significance of LTI systems ✓
 - Examples
 - DFT, DTFT, Z-transform
- Causal system example: The Golconda fort security system.
- Non-causal system example: Bidirectional prediction in a video encoder.
- Stable systems in the BIBO sense.
 - Differencing system is BIBO stable
 - Accumulator system is not BIBO stable. (e.g., $x[n] = u[n]$)
- Linear time invariant ^(LTI) systems: An important class of discrete time systems
 - A system $T\{\}$ is LTI if it is linear and time-invariant.

Q. Where does this lead us?

- A. Recall that any discrete-time signal $x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$
- Let us analyse what happens when we input $x[n]$ to an LTI $T\{\}$.
- Since -
- $$\delta[n] = \begin{cases} 1, & n=0 \\ 0, & n \neq 0 \end{cases}$$
- $$\delta[n-k] = \begin{cases} 1, & n=k \\ 0, & n \neq k \end{cases}$$

$$y[n] = T\{x[n]\}$$

$$= T\left\{ \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \right\}$$

(from ①)

$$= \sum_{k=-\infty}^{\infty} T\{x[k] \cdot \delta[n-k]\}$$

$$= \sum_{k=-\infty}^{\infty} x[k] \cdot T\{\delta[n-k]\} \quad \checkmark$$

If $h[n] = T\{\delta[n]\}$, then let's say
 $h_k[n] = T\{\delta[n-k]\}$.

$$= \sum_{k=-\infty}^{\infty} x[k] \cdot h_k[n] \quad - (2)$$

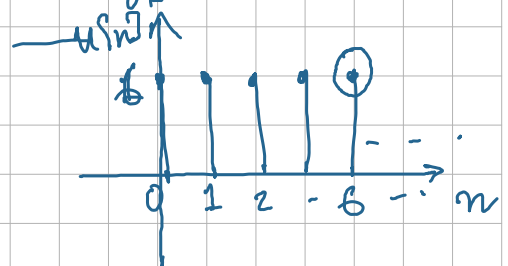
(due to the linearity prop.)
 recall that a linear $T\{\}$ satisfies the following

$$T\left\{\sum_{k=1}^K \alpha_k x_k[n]\right\} =$$

$$\sum_{k=1}^K \alpha_k T\{x_k[n]\} =$$

$$\sum_{k=1}^K \alpha_k y_k[n] \quad \text{where}$$

$$y_k[n] = T\{x_k[n]\}$$



$$x[k] = \underbrace{\delta[n-k]}_{x_k[n]} \quad k \geq 0$$

Recall: If $x[n] = \delta[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$

$$= \sum_{k=-\infty}^n \delta[k]$$

$$= \sum_{k=-\infty}^{\infty} x[k] h[n-k] \quad - (2) \quad (\text{due to time invariance})$$

If $h[n] = T\{\delta[n]\}$

$$h_k[n] = T\{\delta[n-k]\},$$

then $h_k[n] = h[n-k]$.

$$\therefore y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

The convolution sum.

$h[n] = T\{\delta[n]\}$ is the impulse response.

- The expression in (3) is much more convenient for us compared to (2)
- In other words, computing $y[n]$ for an LTI system is easier than computing it for a linear system.
- This is because the LTI system is completely defined by $h[n] = \mathcal{T}\{\delta[n]\}$, unlike a linear system which requires us to know $h_k[n] = \mathcal{T}\{\delta[n-k]\}$ for all k .

• If $y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$, $Y(e^{j\omega}) = X(e^{j\omega}) \cdot H(e^{j\omega})$

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} y[n] e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k] \cdot \underbrace{e^{-j\omega n}}$$

$$= \sum_n \sum_k x[k] \cdot h[n-k] \cdot e^{-j\omega(n-k)} \cdot e^{-j\omega k}$$

$$= \sum_k \underbrace{\left(\sum_n h[n-k] \cdot e^{-j\omega(n-k)} \right)}_{H(e^{j\omega})} \cdot x[k] e^{-j\omega k} = H(e^{j\omega}) \cdot X(e^{j\omega})$$